SpecSIA: Reliable Source Inference in Federated Learning

Current-round, passive, reference-free attribution

1 Inputs

Attack view (passive server)

Target sample z = (x, y)

x

y : cat

Pre-round global model

θ_t

Current-round client updates

$\Delta\theta_t$

Attack uses only **current-round uploads**.
No local data or historical updates.

SpecSIA: 3-Step Pipeline

1 Round-Anchored Contribution Attribution (RCA)

Compare the target gradient with each current-round update from the same pre-round model.

Compare

Target gradient g_z

Client updates $\{\Delta\theta_i\}$

2 Adaptive Spectral Filtering (ASF)

Suppress shared directions and retain client-specific evidence.

Shared signals

Client-specific evidence

3 Source Attribution

Rank clients and predict the source.

Predicted source client $\hat{\imath}$

1 2 3 ... K

Evaluation (integrated)

Mean ASR
Overall rounds success rate.

Rank-based comparison
Cross-setting relative performance (e.g., mean rank / rank stability).

Cross-round stability
Consistency across communication rounds.

◎ Goal: more accurate and more reliable source attribution.